Lateral Augmentation Using a Non-Resorbable Titanium-Reinforced PTFE Membrane

Clinical Case Report

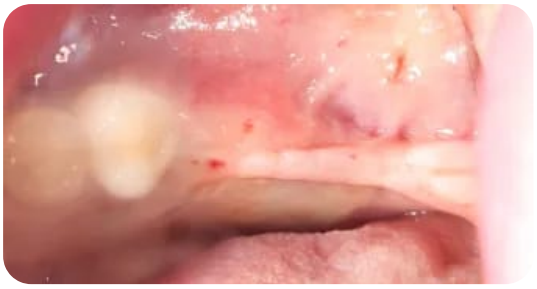
Graft Material Used: 50% autogenous bone + 50% Ivory graft
Application: Lateral augmentation prior to implant placement
Performed by: Dr. Avi Kuperschlag



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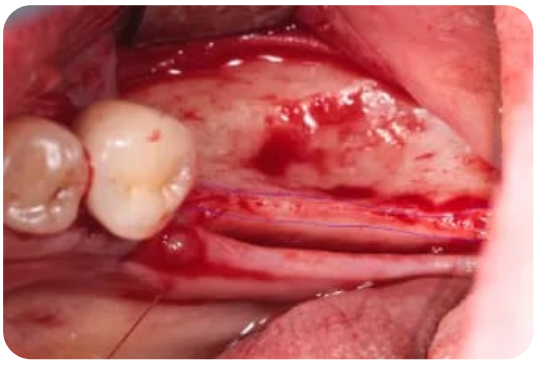
1 Pre-operative condition

A clinical view demonstrating the edentulous area (tooth 36 region). There appears to be insufficient alveolar ridge width for dental implant placement, indicating the necessity of lateral bone augmentation.



2 Intra-operative

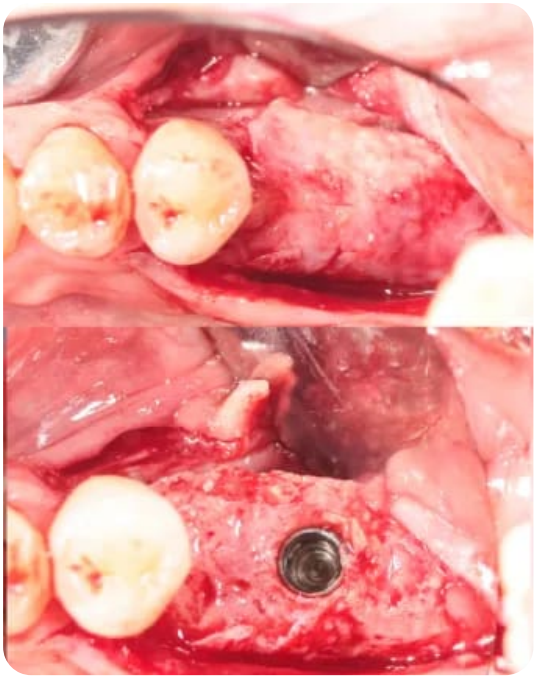
Reflection of the mucoperiosteal flap clearly reveals significant horizontal bone deficiency and exposure of the alveolar ridge. Bleeding from the bone indicates good vascularity, essential for successful grafting.



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3 Implant Placement

Four months post-augmentation, a reopened surgical site shows excellent bone regeneration, providing sufficient volume for the placement of a dental implant. The healing abutment is inserted, and good initial stability appears to have been achieved.



4 Comparison of pre- and post-flap elevation

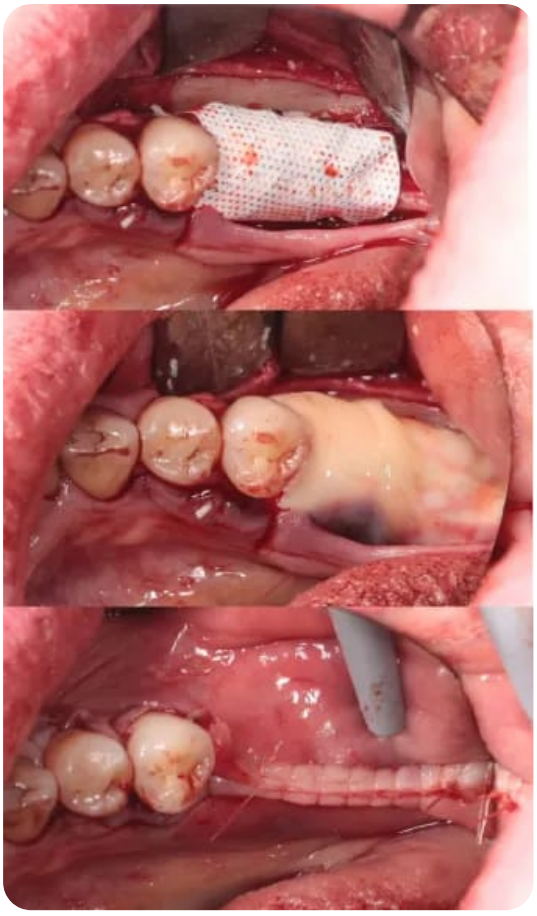
Pre-operative appearance vs. initial exposure of the bone defect after flap elevation. A significant lack of bone structure is evident, justifying the decision for augmentation.



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5 Membrane placement and coverage

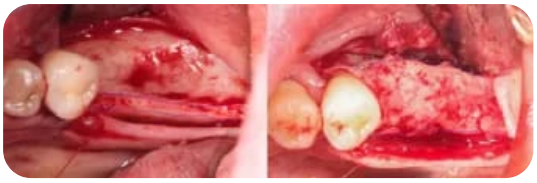
Titanium-reinforced PTFE membrane in position covering the graft site. Two LPRF membranes placed to enhance healing and minimize complications. Site closure appears complete with tension-free flap management and secure suturing.



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6 Direct comparison of the alveolar ridge pre-augmentation and immediately post-graft placement

Clearly shows the augmented bone thickness, demonstrating a successful lateral ridge augmentation with graft materials.



Case Outcome:

After a healing period of only four months, a successful reopening procedure showed excellent bone regeneration at the grafted site, which enabled optimal implant placement and subsequent prosthetic rehabilitation.

This case highlights the efficacy and potential clinical benefits of combining autogenous bone with Ivory Dentin Graft™ materials, supported by a titanium-reinforced PTFE membrane and enhanced by LPRF membranes, achieving predictable results in complex alveolar ridge deficiencies.